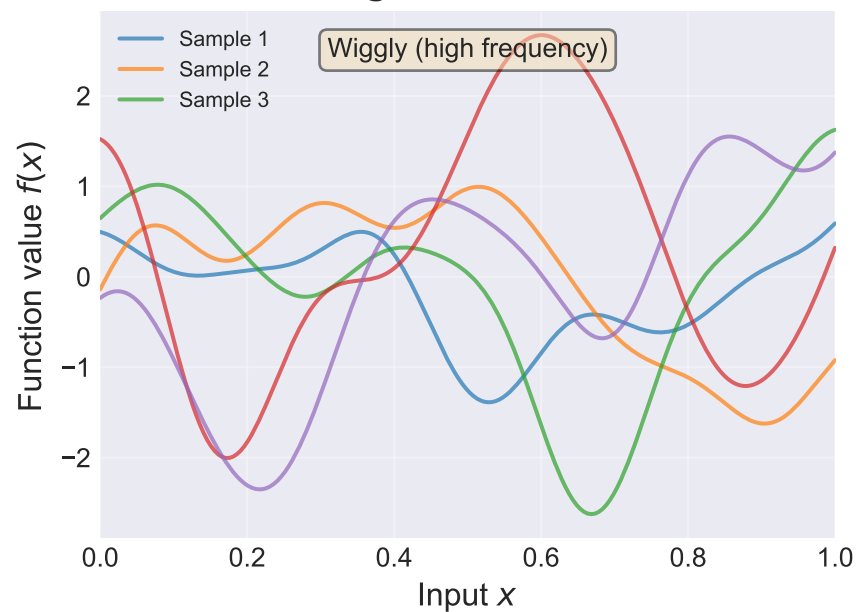
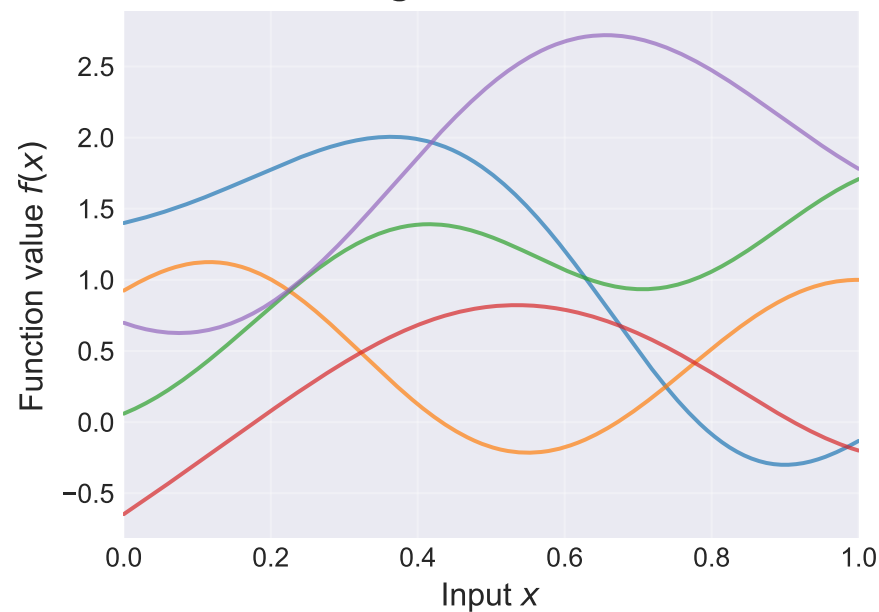


GP Prior Samples: How Lengthscale Controls Function Smoothness

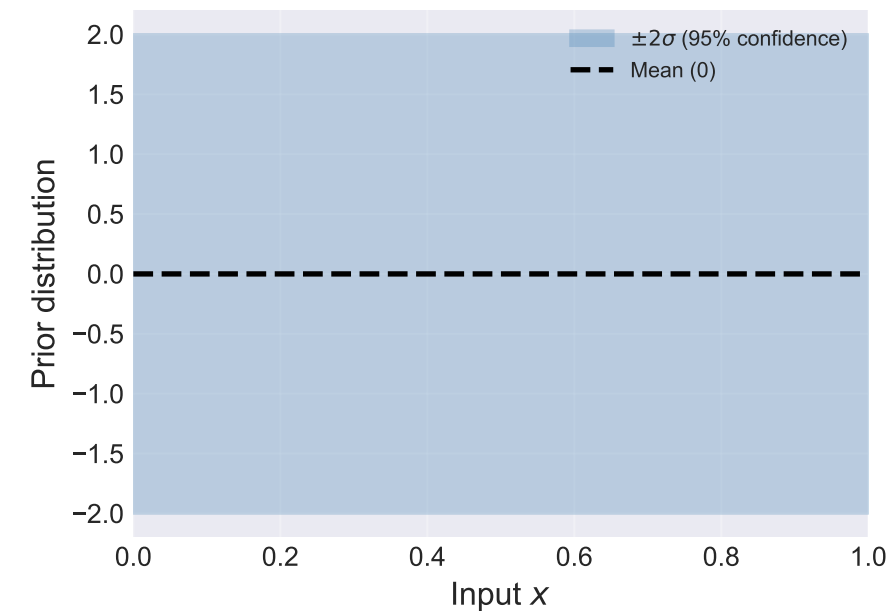
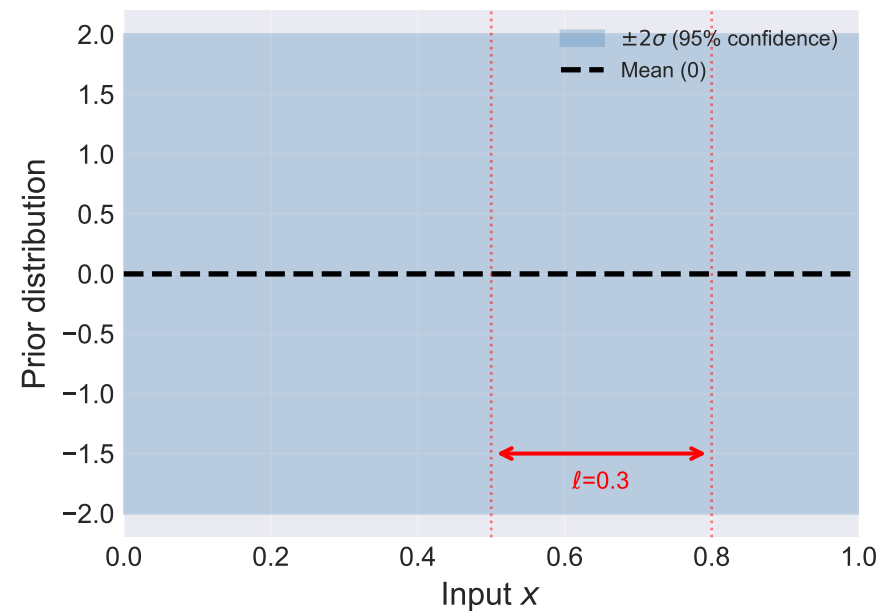
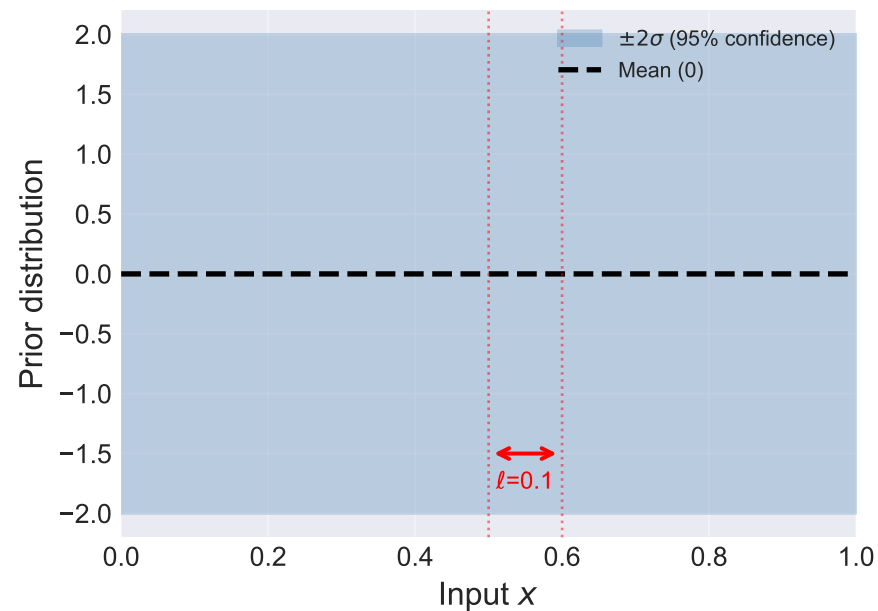
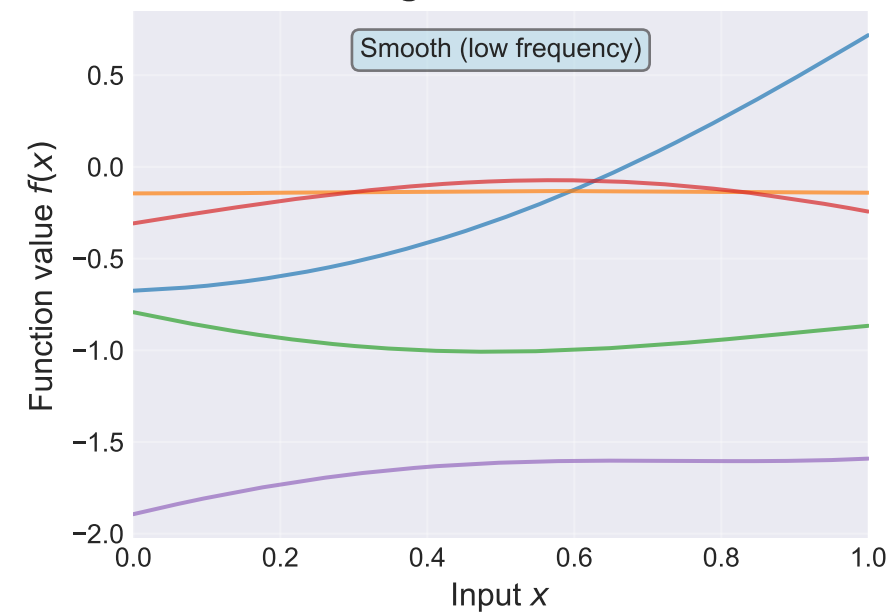
Lengthscale $\ell = 0.1$



Lengthscale $\ell = 0.3$



Lengthscale $\ell = 1.0$



Key Insight: Small lengthscales $\rightarrow$ functions vary rapidly (need dense training data). Large lengthscales $\rightarrow$ functions vary slowly (sparse data sufficient).